STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-LIFT-EQUIVALENCE-MODULE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2212-2235 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 2237-2290, the complete author proof (52 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 7 of queue_examples-defos.json (pb=2202, stmt 2212-2235, proof 2237-2290).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0DYR: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2290–2290. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DZ1: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2170–2181. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0A6A: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2290–2290. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05EC: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2290–2290. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Consider a commutative diagram of Noetherian rings*

$$\begin{array}{ccccc} A' & \longrightarrow & P' & \longrightarrow & Q' \\ \downarrow & & \downarrow & & \downarrow \\ A & \longrightarrow & P & \longrightarrow & Q \end{array}$$

Date: stacks-project:examples-defos.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

with cartesian squares, with flat horizontal arrows, and with surjective vertical arrows whose kernels are nilpotent. Let $J' \subset P'$ be an ideal such that $P'/J' = Q'/J'Q'$. Let M be an A-flat P-module. Assume for all $g \in J'$ there exists an A'-flat $(P')_g$-module lifting M_g. Then the following are equivalent

(1) *M has an A'-flat lift to a P'-module, and*
(2) *$M \otimes_P Q$ has an A'-flat lift to a Q'-module.*

Proof. Let $I = \mathrm{Ker}(A' \to A)$. By induction on the integer $n > 1$ such that $I^n = 0$ we reduce to the case where I is an ideal of square zero; details omitted. We translate the condition of liftability of M into the problem of finding an object of $D^-(P')$ as in Lemma 0DZ1. The obstruction to doing this is the element

$$\omega(M) \in \mathrm{Ext}^2_P(M, M \otimes^{\mathbf{L}}_P IP) = \mathrm{Ext}^2_P(M, M \otimes_P IP)$$

constructed in Deformation Theory, Lemma 0DYR. The equality in the displayed formula holds as $M \otimes^{\mathbf{L}}_P IP = M \otimes_P IP$ since M and P are A-flat[1]. The obstruction for lifting $M \otimes_P Q$ is similarly the element

$$\omega(M \otimes_P Q) \in \mathrm{Ext}^2_Q(M \otimes_P Q, (M \otimes_P Q) \otimes_Q IQ)$$

which is the image of $\omega(M)$ by the functoriality of the construction $\omega(-)$ of Deformation Theory, Lemma 0DYR. By More on Algebra, Lemma 0A6A we have

$$\mathrm{Ext}^2_Q(M \otimes_P Q, (M \otimes_P Q) \otimes_Q IQ) = \mathrm{Ext}^2_P(M, M \otimes_P IP) \otimes_P Q$$

here we use that P is Noetherian and M finite. Our assumption on $P' \to Q'$ guarantees that for an P-module E the map $E \to E \otimes_P Q$ is bijective on J'-power torsion, see More on Algebra, Lemma 05EC. Thus we conclude that it suffices to show $\omega(M)$ is J'-power torsion. In other words, it suffices to show that $\omega(M)$ dies in

$$\mathrm{Ext}^2_P(M, M \otimes_P IP)_g = \mathrm{Ext}^2_{P_g}(M_g, M_g \otimes_{P_g} IP_g)$$

for all $g \in J'$. Howeover, by the compatibility of formation of $\omega(M)$ with base change again, we conclude that this is true as M_g is assumed to have a lift (of course you have to use the whole string of equivalences again). □

[1]Choose a resolution $F_\bullet \to I$ by free A-modules. Since $A \to P$ is flat, $P \otimes_A F_\bullet$ is a free resolution of IP. Hence $M \otimes^{\mathbf{L}}_P IP$ is represented by $M \otimes_P P \otimes_A F_\bullet = M \otimes_A F_\bullet$. This only has cohomology in degree 0 as M is A-flat.